NO TEMPERATURE CONTROL WITH CLIMATE CONTROL UNIT [FULL-AUTO AIR CONDITIONER]

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DESCRIPTION	• Temperature does not change when operating temperature dial • Malfunction in A/C unit and/or climate control unit air mix system • Malfunction in driver and/or passenger side air mix actuator or related wiring harness
POSSIBLE CAUSE	• INTERIOR 15 A fuse malfunction • Open or short circuit in wiring harness between fuse block and climate control unit • Open or short circuit in wiring harness between climate control unit and air mix actuator • Air mix actuator malfunction • Air mix link malfunction • Air mix door malfunction • Climate control unit malfunction • Heater hose malfunction

- When performing an asterisked (*) troubleshooting inspection, shake the wiring harness and connectors while performing the inspection to discover whether poor contact points are the cause of any intermittent malfunctions. If there is a problem, check to make sure connectors, terminals and wiring harnesses are connected correctly and undamaged.

Diagnostic Procedure

Step	Inspection		Action
1	**INSPECT COOLANT TEMPERATURE** • Is the coolant sufficiently warmed up?	Yes	Go to the next step.
		No	Warm up the engine, then go to Step 19.
2*	**INSPECT FUSE** • Remove the INTERIOR 15 A fuse. • Is the fuse normal?	Yes	Go to the next step.
		No	**If the fuse is melted:** Repair or replace wiring harness for short to ground and replace the fuse. **If the fuse is deterioration:** Replace the fuse. Then go to Step 19.
3*	**INSPECT TO SEE WHETHER MALFUNCTION (OPEN CIRCUIT) IS IN B+ SIGNAL WIRING HARNESS (BETWEEN FUSE BLOCK AND CLIMATE CONTROL UNIT) OR ELSEWHERE** • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Switch the ignition ON (engine off or on). • Measure voltage at the following terminal of climate control unit. — Terminal L • Is the voltage **approx. 12 V?**	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between fuse block terminal and climate control unit terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to Step 19.
4*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR AIR MIX ACTUATOR** • Switch the ignition off. • Reconnect the disconnected climate control unit connector. • Switch the ignition ON (engine off or on). • Measure voltage at the following terminals of climate control unit when temperature setting to MAX HOT and MAX COLD. — Terminal G • Is voltage normal?	Yes	Go to Step 15.
		No	Go to the next step.
5*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN POSITION SENSOR OR ELSEWHERE** • Measure voltage at the following terminal of climate control unit when airflow mode at VENT and DEFROSTER. — Terminal I • Is voltage normal?	Yes	Go to Step 8.
		No	Go to the next step.

Step	Inspection		Action
6*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY OR SHORT TO BATTERY POWER OR GROUND) IS IN POSITION SENSOR POWER SUPPLY OR ELSEWHERE** • Measure voltage at the following terminal of climate control unit. — Terminal H • Is voltage **approx. 5 V?**	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
7*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY OR OPEN) IS IN POSITION SENSOR GROUND OR ELSEWHERE** • Measure voltage at the following terminal of climate control unit. — Terminal E • Is the voltage **below. 1.0 V?**	Yes	Go to Step 11.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
8	**INSPECT TO SEE WHETHER MALFUNCTION (SHORT TO POWER SUPPLY) IS IN WIRING HARNESS (AIR MIX POSITION SIGNAL) OR ELSEWHERE** • Measure voltage at the following terminals of climate control unit. — Terminal G • Is the voltage **approx. 12V,** at Step 4?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 19.
		No	Go to the next step.
9*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN WIRING HARNESS (AIR MIX POSITION SIGNAL) OR ELSEWHERE** • Switch the ignition off. • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Disconnect the air mix actuator connector. (See AIR MIX ACTUATOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Verify continuity at following terminals between climate control unit and air mix actuator. — Terminal G —Terminal D • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and each side air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.

Step	Inspection		Action
10*	**INSPECT TO SEE WHETHER MALFUNCTION (SHORT TO GROUND) IS IN WIRING HARNESS (AIR MIX POSITION SIGNAL) OR POSITION SENSOR COMMON (POWER SUPPLY OR GROUND) HARNESS** • Climate control unit and air mix actuator connectors are disconnected. • Verify continuity at the following terminals between climate control unit and ground. — Terminal G • Is there continuity?	Yes	Repair wiring harness for short to ground between climate control unit and air mix actuator. Then go to Step 19.
		No	Inspect and repair for open circuit following: • Between driver side air mix actuator terminal G and junction point to airflow mode actuator (position sensor power supply). • Between passenger side air mix actuator terminal E and junction point to airflow mode actuator (position sensor power supply). • Between driver side air mix actuator terminal E and junction point to airflow mode actuator (position sensor ground). • Between passenger side air mix actuator terminal G and junction point to airflow mode actuator (position sensor ground). Then go to Step 19.
11*	**INSPECT TO SEE WHETHER MALFUNCTION (LACK OF CONTINUITY) IS IN AIR MIX ACTUATOR OR WIRING HARNESS (BETWEEN CLIMATE CONTROL UNIT AND AIR MIX ACTUATOR)** • Disconnect the air mix actuator connector. (See AIR MIX ACTUATOR REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Switch the ignition ON (engine off or on). • Measure voltage at the following terminals of air mix actuator. — Terminal W (COLD motor drive signal) — Terminal X (HOT motor drive signal) • Are voltages as shown below? — COLD motor drive signal: **approx. 10 V** during COLD and **below 1.0 V** during HOT — HOT motor drive signal: **below 1.0 V** during COLD and **approx. 10 V** during HOT	Yes	Go to Step 15.
		No	Go to the next step.
12*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO B+ BETWEEN CLIMATE CONTROL UNIT AND AIR MIX ACTUATOR) OR ELSEWHERE** • Air mix actuator connectors are disconnected. • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Switch the ignition ON (engine off or on). • Measure voltage at the following terminals of the climate control unit harness-side connector. — Terminal W (COLD motor drive signal) — Terminal X (HOT motor drive signal) • Are voltages **approx. 0 V?**	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to power supply. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to power supply. Go to Step 19.
13*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (SHORT TO GROUND BETWEEN CLIMATE CONTROL UNIT AND AIR MIX ACTUATOR) OR ELSEWHERE** • Climate control unit and air mix actuator connectors are disconnected. • Switch the ignition off. • Verify that continuity at the following harness-side connector terminals between climate control unit and ground. — Terminal W (COLD motor drive signal) — Terminal X (HOT motor drive signal) • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to Step 19.
		No	Go to the next step.

Step	Inspection		Action
14*	**INSPECT TO SEE WHETHER MALFUNCTION IS IN WIRING HARNESS (LACK OF CONTINUITY OR OPEN BETWEEN CLIMATE CONTROL UNIT AND AIR MIX ACTUATOR) OR ELSEWHERE** • Climate control unit and air mix actuator connectors are disconnected. • Switch the ignition off. • Verify that continuity at the following harness-side connector terminals between climate control unit and each air mix actuator. • Terminal B (L.H.D.)/A (R.H.D.)—climate control unit terminal W (COLD motor drive signal) • Terminal A (L.H.D.)/B (R.H.D.)—climate control unit terminal X (HOT motor drive signal) • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and air mix actuator terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness. Go to Step 19.
15	**INSPECT AIR MIX ACTUATOR** • Inspect the air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is air mix actuator normal?	Yes	Go to the next step.
		No	Replace the suspected side air mix actuator. Then go to 19. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
16	**INSPECT AIR MIX LINK** • Inspect the air mix links for each-side. (See A/C UNIT REMOVAL/INSTALLATION.) — Is grease on link? — Are links securely and properly installed? — Are links free of obstructions and hindrances? • Are the above items normal?	Yes	Go to the next step.
		No	Apply grease to links. If any links are damaged. Replace malfunctioning part. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) Then go to Step 19.
17	**INSPECT TO SEE WHETHER MALFUNCTION IS IN CLIMATE CONTROL UNIT OR AIR MIX DOOR ELSEWHERE** • Inspect the A/C unit and each-side air mix door. (See A/C UNIT REMOVAL/INSTALLATION.) — Is door free of obstructions and damage? — Are doors securely and properly installed? • Are the above items normal?	Yes	Go to the next step.
		No	Remove the obstruction, or install door in proper position. If any door are cracked, replace them. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) Then go to the next step.
18	**INSPECT HEATER HOSE** • Inspect the heater hoses for the following: — Is heater hoses free of damage and cracks? — Are heater hoses connections free of engine coolant leakage? — Are heater hoses installation points on A/C unit free of engine coolant leakage? • Are the above items normal?	Yes	Replace the climate control unit. Then go to the next step. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].)
		No	If heater hoses connections are loose, install securely in the correct position. Repair or replace the heater hoses. (See HEATER HOSE REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See HEATER HOSE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].) Then go to the next step.
19	**VERIFY THAT MALFUNCTION SYMPTOM DOES NOT RECUR AFTER REPAIR** • Does the unit operate at every temperature setting?	Yes	Troubleshooting completed. Explain repairs to customer.
		No	Recheck malfunction symptoms, then repeat from Step1 if the malfunction recurs.